Spatial and Temporal Analysis of Assists in Football
Case Study: Lorenzo Insigne → José Callejón (2017/18 Season)

Data Science Project Applied to Sports Analytics

1 Introduction

In recent years, data analysis has become a central component of professional football, enabling a deeper understanding of tactical behaviors, player interactions, and offensive efficiency. This project fits within the field of *football analytics*, with the aim of analyzing assist and finishing dynamics through a spatial representation of match events.

The case study focuses on the 2017/18 season, analyzing the offensive connection between Lorenzo Insigne and José Callejón, two key players in Napoli's attacking system.

2 Objective of the Analysis

The main objectives of the analysis are:

- to study the spatial distribution of assists provided by Insigne to Callejón;
- to distinguish between action outcomes (goals and non-goal shots);
- to identify recurring patterns in creation and finishing zones;
- to evaluate the effectiveness of the Insigne–Callejón partnership in offensive production.

The analysis goes beyond simple event counting and aims to provide a qualitative tactical interpretation through data visualization.

3 Dataset and Methodology

The dataset includes offensive events from the 2017/18 season, with particular focus on actions where Insigne is the final passer and Callejón is the shooter.

For each event, the following information was considered:

- assist origin coordinates;
- shot location coordinates;

- shot outcome (goal or non-goal).

All spatial data were normalized according to standard pitch dimensions, ensuring a consistent and comparable representation of events.

4 Visualization and Spatial Analysis

Figure 1 shows the spatial distribution of the analyzed actions, overlaid on a standardized football pitch.

- Blue lines represent assists delivered by Insigne.
- Yellow markers indicate shots resulting in goals.
- Red markers represent non-converted shots.

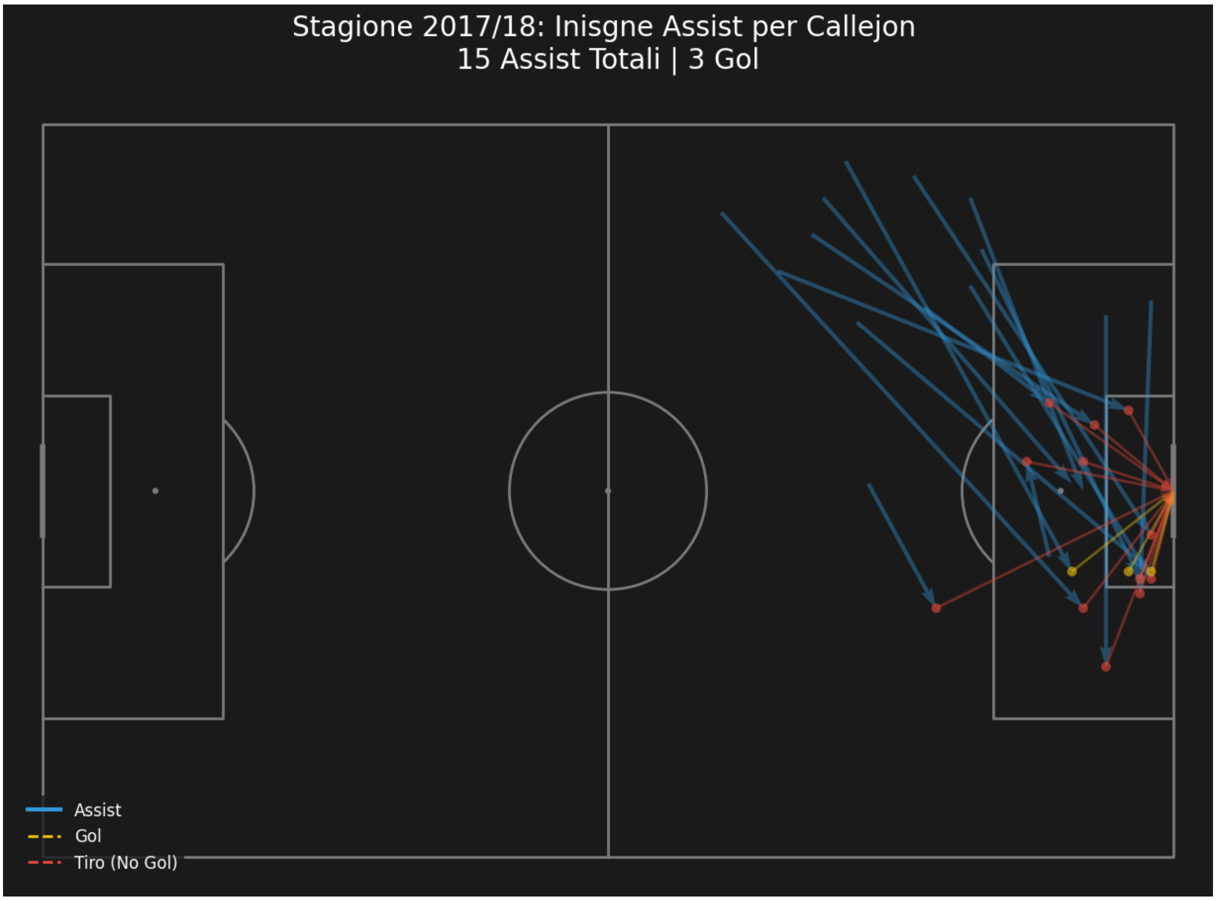

Figure 1: Spatial distribution of Insigne's assists to Callejón – 2017/18 season

The analysis highlights a strong concentration of actions on the right side of the opponent's penalty area, which is consistent with Callejón's typical attacking movement, often aimed at exploiting space at the far post.

5 Results

Overall, the analysis shows:

- 15 total assists from Insigne to Callejón;
- 3 goals scored following these assists;
- a prevalence of assists originating from the left half-space;
- a high repetition of offensive patterns, indicating well-established tactical automatisms.

The assist-to-goal ratio suggests strong offensive output, with room for improvement in terms of shot conversion.

6 Tactical Interpretation

From a tactical perspective, the results confirm:

- Insigne's role as an advanced playmaker;
- the importance of Callejón's off-ball movements;
- the effectiveness of attacking rotations within the adopted playing system.

The repeated passing trajectories clearly indicate a structured and rehearsed attacking pattern, consistently exploited throughout the season.

7 Conclusions

This study demonstrates how spatial data analysis can provide valuable insights into a team's offensive behavior and player relationships.

The proposed approach is easily extendable to other contexts, such as comparisons across seasons, teams, or different player pairs, making it a useful tool for analysts, coaches, and technical staff.

8 Future Work

Potential future developments include:

- integration of *Expected Goals (xG)* metrics;
- temporal analysis of actions (match minute);
- comparison with other wide attackers from the same season;
- application of predictive models to estimate goal probability.